SECTION A : 5 marks each

1. Let a and b be positive integers, prove that the equation $x^2 + (a+b)^2 x + 4ab = 1$ has rational solutions if and only if $a = b$.

Solution : ($\Leftarrow$) *if $a = b$, the equation becomes*

$$x^2 + 4a^2 x + 4a^2 - 1 = 0$$

and its solutions are:

$$x_1 = -1$$

and

$$x_2 = 1 - 4a^2,$$

which are clearly rational

($\Rightarrow$) *Assume that the equation*

$$x^2 + 5x + p = 0,$$

with

$$s = (a+b)^2$$

and

$$p = 4ab - 1,$$

has rational roots.

Then the discriminant $\Delta = s^2 - 4p$ must be a square, hence

$$s^2 - 4p = n^2$$

for some n a positive integer ($n \in \mathbb{N}$).

Since $4p > 0$ it follows that $n < s$, but s and n have the same parity.

Therefore, $n \leq s - 2$.

We obtain

$$s^2 - 4p \leq (s-2)^2 \Leftrightarrow s \leq p + 1,$$

which is equivalent to

$$(a+b)^2 \leq 4ab.$$

This results in

$$(a-b)^2 = 0,$$

hence

$$a = b$$

■

2. Find all pairs of integers m and n that satisfy $m^2 = 57 + n^2$.

Solution :

$$m^2 - n^2 = 57.$$

That is,

$$(m-n)(m+n) = 3 \cdot 19.$$

Case 1: $m - n = 1$ *and* $m + n = 57$

We obtain $(m, n) = (29, 28)$ *. In fact,* $29^2 - 28^2 = 57$.

Case 2: $m - n = 3$ *and* $m + n = 19$

We obtain $(m, n) = (11, 8)$ *. In fact,* $11^2 - 8^2 = 57$ *.*

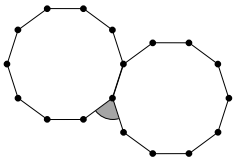
There are two pairs: $(m, n) = (29, 28)$ *and* $(11, 8)$.

■

3. Given an $n \times n \times n$ grid of unit cubes, a cube is good if it is a sub-cube of the grid and has side length at least two. If a good cube contains another good cube and their faces do not intersect, the first good cube is said to *properly* contain the second. What is the size of the largest possible set of good cubes such that no cube in the set properly contains another cube in the set?

Solution : *Let S be the set of good cubes with side length 2 or 3. For any $s \in S$, we define C_s to be the set of good cubes that have s as their centre. Then we can see that for any s and any pair of distinct cubes in C_s, that one cube properly contains the other. Also, for any good cube c_1 in the grid, there exists a cube $c_2 \in S$ such that c_2 is the centre of c_1. Thus, the size of largest possible set of good cubes with no cube properly containing each other is at most $|S|$. Finally, we observe that no cube in S can properly contain another cube in S. Thus $|S| = (n-1)^3 + (n-2)^3$ is the size of the largest possible set.* ■

4. This floor plan of the portal to the Addams castle shows twin chambers and a wedge shaped pit (shaded) where a trap door will be placed. Find the degree measure of the acute angle at the tip of the wedge. The chambers are congruent regular ten-sided polygons.



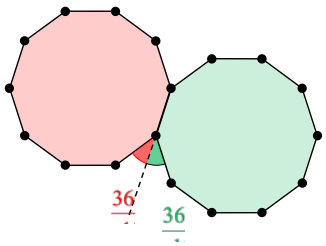
Solution 1 : *One exterior angle of a regular n-sided polygon measures*

$$\frac{360^\circ}{n}$$

*One exterior angle of a regular decagon (*10*-sided polygon) measures*

$$\frac{360^\circ}{10} = 36^\circ.$$

The marked angle contains two exterior angles of regular decagons as shown.



The marked angle in the original diagram measures

$$2 \cdot 36^\circ = 72^\circ.$$

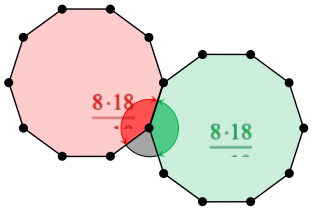
■

Solution 2 : *One interior angle of a regular n-sided polygon measures*

$$\frac{(n-2) \cdot 180^\circ}{n}$$

One interior angle of a regular decagon measures

$$\frac{8 \cdot 180^\circ}{10} = 144^\circ$$



The marked angle in the original diagram measures

$$360^\circ - 2 \cdot 144^\circ = 72^\circ$$

■

5. What is the sum of the digits of the integer equal to $3 \times 10^{500} - 2022 \times 10^{497} - 2022$?

Solution :

$$\begin{aligned} 3 \times 10^{500} - 2022 \times 10^{497} - 2022 &= 3000 \times 10^{497} - 2022 \times 10^{497} - 2022 \\ &= (3000 - 2022) \times 10^{497} - 2022 \\ &= 978 \times 10^{497} - 2022 \\ &= (978 \times 10^{497} - 1) - 2021 \end{aligned}$$

The integer $978 \times 10^{497} - 1$ is the 500-digit integer with leading digits 977 followed by 497 $9's$. Thus, the digits of the given integer, from left to right, are 9, 7, and 7 followed by $497 - 4 = 493$ $9's$ and the final four digits are $9 - 2 = 7$, $9 - 0 = 9$, $9 - 2 = 7$, and $9 - 1 = 8$. Therefore, the sum of the digits of the integer is

$$9 + 7 + 7 + (493 \times 9) + 7 + 9 + 7 + 8 = 4491.$$

■

6. Suppose $f(x) = x^2 + 12x + 30$.

   Find all real numbers x for which $f(f(f(f(f(x))))) = 0$.

Solution :

$$f(x) = (x+6)^2 - 6.$$

So

$$\begin{aligned} f(f(x)) &= f((x+6)^2 - 6) \\ &= (x+6)^4 - 6, \\ &\dots \\ f(f(f(f(f(x))))) &= (x+6)^{32} - 6. \end{aligned}$$

The equation becomes

$$x + 6^{32} = 6.$$

The 32 *roots are*

$$x = -6 + \sqrt[32]{6}e^{\frac{2k\pi}{32}i}$$

for

$$k = 0, 1, 2, \dots, 31.$$

We have only two real roots when $k = 0$ *and* $k = 16$ *(the regular* 32*-gon has only two vertices on the* x*-axis):*

$$x = -6 + \sqrt[32]{6}$$

and

$$x = -6 - \sqrt[32]{6}.$$

■

7. Let $a_0, a_1, a_3, \dots$ be a be a sequence of integers (positive, negative, or zero) such that for all nonnegative integers n and k,

$$a_{n+k}^2 - (2k+1)a_n a_{n+k} + (k^2+k)a_n^2 = k^2 - k.$$

Find all possible sequence a_n.

Solution 1 : *We can factor the equation as*

$$\big[a_{n+k} - ka_n\big]\big[a_{n+k} - (k+1)a_n\big] = k(k-1). \tag{1.1}$$

Substituting in $k = 1$*, we find that either* $a_{n+1} = 2a_n$ *or* $a_{n+1} = a_n$ *for all integers* n*. If* p *is a prime such that* $p \mid a_r$*, then by repeatedly applying* $a_{n+1} = 2a_n$ *or* $a_{n+1} = a_n$*, we find that* p *divides* a_n *for all integers* $n \geq r$*. Substituting* $k = 2$ *and* $n = r$ *into (1.1) we notice that the LHS is divisible by* p^2*, and the RHS is equal to* 2*; hence* $p^2 \mid 2$*, which is a contradiction. Thus* a_n *is never divisible by any prime for any* n*, so* $a_n = \pm 1$*.*

In particular, $a_{n+1} = a_n$ *for all integers* n*, and* a_n *is constant-either* 1 *or* -1*. Substituting either of these sequences into (1.1) shows that both are valid solutions.* ■

Solution 2 : *After multiplying by 4 and completing the square twice, we find*

$$2a_{n+k} - (2k+1)a_n)^2 - (2k-1)^2 = a_n^2 - 1.$$

As the closest square to $(2k-1)^2$ is $(2k-2)^2$ (for k positive), we see that if $x \neq 2k-1$ is an integer, then $\mid x^2 - (2k-1)^2 \mid \geq 4k-3$. Thus if $\mid a_n^2 - 1 \mid \neq 0$, then $\mid a_n^2 - 1 \mid \geq 4k-3$ for all positive integers k. But this contradicts the fact that $\mid a_n^2 - 1 \mid$ is finite. Therefore, $a_n^2 = 1$, and $\mid 2a_{n+k} - (2k+1)a_n \mid = 2k-1$. Now if $a_{n+k} = -a_n$, then $\mid 2a_{n+k} - (2k+1)a_n \mid = 2k+3$, which is not equal to $2k-1$, hence $a_{n+k} \neq -a_n$. But as $a_n^2 = 1$ for all n, we see that either $a_n = 1$ for all n, or $a_n = -1$ for all n. It is easy to check that both of these sequences work. ■

Solution 3 : *Factoring the equation as in (1.1) with $k = 1$, we find that either $a_{n+1} = a_n$ or $a_{n+1} = 2a_n$ for all nonnegative integers n. It follows that, for all nonnegative integers n, either*

(i) $a_{n+2} = a_n$, *or*
(ii) $a_{n+2} = 2a_n$, *or*
(iii) $a_{n+2} = 4a_n$.

Now factor the equation with $k = 2$, and we get

$$(a_{n+2} - 2a_n)(a_{n+2} - 3a_n) = 2.$$

Fix $n \geq 0$. Consider the cases (7i), (7ii), and (7iii). In case (7ii), the above gives $0 = 2$, a contradiction. In case (7i), the above gives $(-a_n)(-2a_n) = 2$, so, $a_n = 1$. In case (7iii), $(2a_n)(a_n) = 2$. In any case, $a_n^2 = 1$, so that $a_n = \{-1, 1\}$ for all n.
Because a_{n+1} has to be either a_n or $2a_n$, and $a_n \in \{-1, 1\}$ for all n, a_n must be constant. Thus the only sequences that satisfy the desired property are $a_n = 1$ for all $n \geq 0$, and $a_n = -1$ for all $n \geq 0$. ■

8. Find the smallest number of nightshade seedlings Uncle Fester could have if when he arranges them with 8 in each row, he has three left over and when he puts 9 in each row, he has 4 left over.

Solution 1 : *We are searching the number n:*

$$n = 3 = -5 \mod 8,$$
$$n = 4 = -5 \mod 9.$$

The smallest positive value of n is

$$lcm(8, 9) - 5 = 72 - 5 = 67$$

■

Solution 2 :

We are searching the number n:

$$n = 3 \mod 8,$$
$$n = 4 \mod 9.$$

If you don't see the same "negative" remainders in solution 8, we may do the following.

Let $n = 8k + 3$ for some nonnegative integer k. We have

$$8k + 3 = 4 \mod 9.$$

Do mod calculations:

$$8k = 1 \mod 9 \to -k = 1 \mod 9 \to k = -1 = 8 \mod 9.$$

The smallest nonnegative value of k is 8.
The smallest positive value of n is $n = 8 \cdot 8 + 3 = 67$.
The answer is 67.

■

Solution 3 :

If you don't know the "mod" sign, we may think this way.
Let Uncle Fester borrow 5 nightshade seedlings.
When he arranges the original nightshade seedlings +5 borrowed with 8 in each row, he has none left over and when he puts 9 in each row, he has none left over.
Then the least number of nightshade seedlings is the least common multiple of 8 and 9. It is $8 \cdot 9 = 72$.
Now ask Uncle Fester to return 5 nightshade seedlings.
The answer is $72 - 5 = 67$.

■

Solution 4 :

We list the numbers that yield remainder 3 upon division by 8 until we hit a number that yields remainder 4 upon division by 9:

$$3, 11, 19, 27, 35, 43, 51, 59, 67, \ldots.$$

We see that 67 yields remainder 4 upon division by 9.
The answer is 67.

■

9. Let ABC be a triangle with $|AB| < |AC|$, where $|\cdot|$ denotes length. Suppose D, E, F are points on side BC such that D is the foot of the perpendicular on BC from A, AE is the angle bisector of $\angle BAC$, and F is the midpoint of BC. Further suppose that $\angle BAD = \angle DAE = \angle EAF = \angle FAC$. Determine all possible values of $\angle ABC$.

Solution :

The only possible value is $\angle ABC = 90°$.
Let $\theta = \angle BAD = \angle DAE = \angle EAF = \angle FAC$.
Let O be the circumcentre of ΔABC. Then note that
$\angle BAD = 90 - \angle ABD = 90 - \angle ABC = 90 - \frac{1}{2}\angle AOC = \angle OAC$. Therefore, $\theta = \angle OAC = \angle FAC$.
Therefore, A, F, O are collinear.

Since $FB = FC$ and $OB = OC, F, O$ lie on the perpendicular bisector of BC. If F, O are distinct points, then A is also on the perpendicular bisector of BC. But $|AB| < |AC|$, thus this is impossible. Therefore, $F = O$. Thus, the midpoint of BC is the circumcentre of ΔABC. We conclude that $\angle BAC = 90°$.

■

10. Let $\mathbb{R}^+$ denote the set of positive real numbers. Find all functions $f : \mathbb{R}^+ \to \mathbb{R}^+$ satisfying for all $x, y \in \mathbb{R}^+$ the equality

$$f(xf(y)) = f(xy) + x$$

Solution : *Let*

$$P(x, y) = f(xf(y)) = (f(xy) + x)$$

$$\begin{aligned} P(f(x), y) &= f(f(x)f(y)) = (f(f(x)y) + f(x)) = f(f(x)f(y)) + y + f(x) \\ P(f(y), x) &= f(f(x)f(y)) = f(f(x)f(y)) + x + f(y) = P(f(x), y) \end{aligned}$$

So

$$f(x) - x = f(y) - y = c$$
$$f(x) = x + c$$

Putting it into original equation we get $c = 1$. Therefore $f(x) = x + 1$ is appropriate.

■

SECTION B : 10 marks each

11. The notation $[2, 3, 5, 13]$ is an abbreviation for the fraction

$$2 + \cfrac{1}{3 + \cfrac{1}{5 + \cfrac{1}{13}}}$$

Solve for the positive integer a that satisfies this equation:

$$[a + 1, a, a - 1] - [a - 1, a, a + 1] = [1, 1, 665, 2]$$

Solution : *The equation is*

$$a + 1 + \cfrac{1}{a + \cfrac{1}{a - 1}} - a + 1 - \cfrac{1}{a + \cfrac{1}{a + 1}} = 1 + \cfrac{1}{1 + \cfrac{1}{665 + \cfrac{1}{2}}}$$

Let us simplify:

$$\begin{aligned} \frac{a-1}{a^2 - a + 1} - \frac{a+1}{a^2 + a + 1} &= -\frac{2}{1333}, \\ \frac{a^3 - 1 - (a^3 + 1)}{(a^2 - a + 1)(a^2 + a + 1)} &= -\frac{2}{1333}, \\ \frac{-2}{(a^2 - a + 1)(a^2 + a + 1)} &= -\frac{2}{1333}, \\ (a^2 - a + 1)(a^2 + a + 1) &= 1333 \\ &= 31 \cdot 43. \end{aligned}$$

Note that

$$6^2 - 6 + 1 = 31$$

and

$$6^2 + 6 + 1 = 43$$

We obtain

$$a = 6$$

■

12. The integers a and b have a property that the expression

$$\frac{2n^3 + 3n^2 + an + b}{n^2 + 1}$$

is an integer for every integer n. What is the value of the expression above when $n = 4$?

Solution :

$$\begin{aligned}\frac{2n^3 + 3n^2 + an + b}{n^2 + 1} &= \frac{2n^3 + 2n + 3n^2 + 3 + (a-2)n + (b-3)}{n^2 + 1} \\ &= \frac{2n(n^2+1) + 3(n^2+1) + (a-2)n + (b-3)}{n^2 + 1} \\ &= 2n + 3 + \frac{(a-2)n + (b-3)}{n^2 + 1}\end{aligned}$$

The quantity $2n + 3$ is an integer if n is an integer, so the given expression in n is an integer exactly when $\frac{(a-2)n + (b-3)}{n^2 + 1}$ is an integer.
Suppose that $a - 2 \neq 0$ and consider the function $f(x) = x^2 + 1$, $g(x) = (a-2)x + b - 3$, and $h(x) = f(x) - g(x) = x^2 - (a-2)x + 4 - b$.
The graph of $h(x)$ is a parabola with a positive coefficient of x^2, which means that there are finitely many integers n with the property that $h(n) \leq 0$ (there could be no such integers). Since $h(n) = f(n) - g(n)$, we have shown that there are only finitely many integers n for which $g(n) < f(n)$ is false.
The graph of $g(x)$ is a line that is neither vertical nor horizontal since $a - 2 \neq 0$, so there must be infinitely many integers for which $g(n) > 0$.
Since infinitely many integers satisfy $0 < g(n)$ and only finitely many integers fail $g(n) < f(n)$, there must exist an integer n with the property that $0 < g(n) < f(n)$.
This means there exists an integer n for which

$$0 < \frac{g(n)}{f(n)} < 1$$

but

$$\frac{g(n)}{f(n)} = \frac{(a-2)n + b - 3}{n^2 + 1}$$

is not an integer. Therefore, we have shown that if $a - 2 \neq 2$, then there is an integer n for which the expression in the problem is not an integer, so we conclude that $a - 2 = 0$ or $a = 2$. We now have that the expression in the question is equivalent to

$$2n + 3 + \frac{b-3}{n^2 + 1}$$

which is an integer exactly when $\dfrac{b-3}{n^2+1}$ *ia an integer.*

Similarly, if $b-3$ *is negative, then there are integers* n *for which* $\dfrac{b-3}{n^2+1}$ *is not an integer.*

This shows that if $b-3 \neq 0$*, then there are integers* n *for which the expression in the question is not an integer.*

We can conclude that $b-3=0$*¡ or* $b=3$*.*

Therefore, we have that $a=2$ *and* $b=3$*, so the expression given in the problem is equivalent to*

$$2n+3+\frac{(2-2)n+(3-3)}{n^2+1} = 2n+3$$

Thus, the answer to the question is $2(4)+3=11$*.* ■

13. Morticia gives a number between 1 and 65 to Lurch, and then Pugsley tries to guess what it is by asking Lurch questions. Pugsley asks first "Is your number greater than 20?" and Lurch answers. Then Pugsley asks "Is your number even or odd?" and Lurch answers. "Hmm," says Pugsley, "Is your number a perfect square?" When Lurch answers, Pugsley says "I think I almost have it. I will know if you tell me whether one of the digits in your number is a 4." Lurch answers and Pugsley says "I have it!" Just then, Morticia looks up and says "Oh Pugsley-I forgot to tell you! Lurch is having a backwards day today. Every answer he gives is wrong." Pugsley furrows his brow. After a minute he says, "Thanks. That is ok. I can still tell what his number is!" What does Pugsley think is Lurch's number?

Solution : *We analyze in two cases.*

Case 1: the answer is "no" to "is your number greater than 20*?"*

The numbers are from 2 *to* 20 *inclusive.*

We divide the numbers into two groups: even and odd

Even:	$2, 4, 6, 8, 10, 12, 14, 16, 18, 20$
Odd:	$3, 5, 7, 9, 11, 13, 15, 17, 19$

There is a digit 4 *only in some even numbers. Hence the answer to "is your number even or odd?" is "even".*

In the even numbers there are 2 *square numbers:* 4 *and* 16*. One number contains a digit* 4 *and the other doesn't.*

In the even numbers there are 8 *non-square numbers:* $2, 6, 8, 10, 12, 14, 18, 20$*. There are* 7 *numbers not containing a digit* 4*.*

Since Pugsley can know the number by knowing whether of one of digits of the number is a 4*, the answer to "is your number a perfect square?" is "yes".*

If the answer is "yes" to "whether one of the digits is a 4*, Lurch's number is* 4*.*

If the answer is "no" to "whether one of the digits is a 4*, Lurch's number is* 16*.*

Now we examine which situation with the opposite answers still allows Pugsley to determine Lurch's number.

We use the opposite answers for all.

The numbers are greater than $20:21$ *to* 63 *inclusive.*

The numbers are odd: $21, 23, 25, 27, 29, \ldots, 47, 49, 51, 53, \ldots, 63$*.*

The numbers are non-squares: $21, 23, 27, 29, \ldots, 47, 51, 53, \ldots, 63$*.*

If the answer is "no" to "whether one of the digits is a 4*, the number can be one of* $21, 23, 27, 29, \ldots, 39, 51, 53, \ldots, 63$*. Pugsley cannot determine the number.*

If the answer is "yes" to "whether one of the digits is a 4*, the number can be one of* $41, 43, 45, 47$*. Pugsley cannot determine the number either.*

Therefore, there is no solution in this case.

Case 2: the answer is "yes" to "is your number greater than 20*?"*

The numbers are from 21 *to* 63 *inclusive.*

We divide the numbers into two groups: even and odd

$$\begin{aligned} \textit{Even:} &\quad 22, 24, 26, 28, \ldots, 62 \\ \textit{Odd:} &\quad 21, 23, 25, 27, \ldots, 63 \end{aligned}$$

Subcase a: to "is your number even or odd?", the answer is "even"

Since there is only one square number: 36*, the answer to "is your number a perfect square?" is "no".*

Then Pugsley cannot determine Lurch's number because there are two or more numbers containing a digit 4 *and there are two or more numbers not containing a digit* 4 *in the nonsquares.*

Subcase b: to "is your number even or odd?", the answer is "odd"

There are two squares 25 *and* 49*. One contains a digit* 4*, and the other doesn't. Therefore, the answer to "is your number a perfect square?" is "yes".*

If the answer is "yes" to "whether one of the digits is a 4*?", the number is* 49*.*

If the answer is "no" to "whether one of the digits is a 4*?", the number is* 25*.*

Now we examine which situation with the opposite answers still allows Pugsley to determine Lurch's number.

We use the opposite answers for all.

The numbers are less than or equal to $20 : 2$ *to* 20 *inclusive.*

The numbers are even: $2, 4, 6, 8, 10, 12, 14, 16, 18$*.*

The numbers are non-squares: $2, 6, 8, 10, 12, 14, 18$*.*

If the answer is "no" to "whether one of the digits is a 4*?", the number cannot be determined.*

If the answer is "yes" to "whether one of the digits is a 4*?", the number is* 14*. Pugsley can still determine Lurch's number.*

Therefore, Lurch's number is 14*.*

Let us make a summary:

Lurch's number is 14*.*

To "is your number greater than 20*?", the liar's answer is "yes" (the true answer is "no").*

To "is your number even or odd?", the liar's answer is "odd" (the true answer is "even").

To "is your number a perfect square?", the liar's answer is "yes" (the true answer is "no").
To "whether one of the digits is a 4?", the liar's answer is "no" (the true answer is "yes").
The answer to the problem is 14.

■

14. Find $x \in \left(0, \frac{3}{4}\right)$ so that

$$\log_x(1-x) + \log_2 \frac{1-x}{x} = \frac{1}{(\log_2 x)^2}.$$

Solution :

Let

$$a = \log_2 x$$

and

$$b = \log_2(1-x).$$

Then

$$\frac{b}{a} + b - a = \frac{1}{a^2}$$

hence

$$(a+1)(a^2 - a - ab + 1) = 0$$

If $a^2 - a - ab + 1 = 0$, *then* $a \neq 0$ *and*

$$\begin{aligned} b &= \frac{a^2 - a + 1}{a} \\ &= a + \frac{1}{a} - 1, \end{aligned}$$

yielding

$$a + \frac{1}{a} = b + 1$$

But $x \in \left(0, \frac{3}{4}\right)$ *gives* $1 - x \in \left(\frac{1}{4}, 1\right)$,

Hence

$$b = \log_2(1-x) \in (-2, 0)$$

Therefore, $a + \frac{1}{a} \in (-1, 1) \rightarrow$ *impossible,*

Since $\left|t + \frac{1}{t}\right| \geq 2$, *for all* $t \neq 0$

This leaves $a + 1 = 0$, *hence*

$$\begin{aligned} \log_2 x &= -1 \\ \Rightarrow \quad x &= 2^{-1} \\ &= \frac{1}{2} \end{aligned}$$

■